

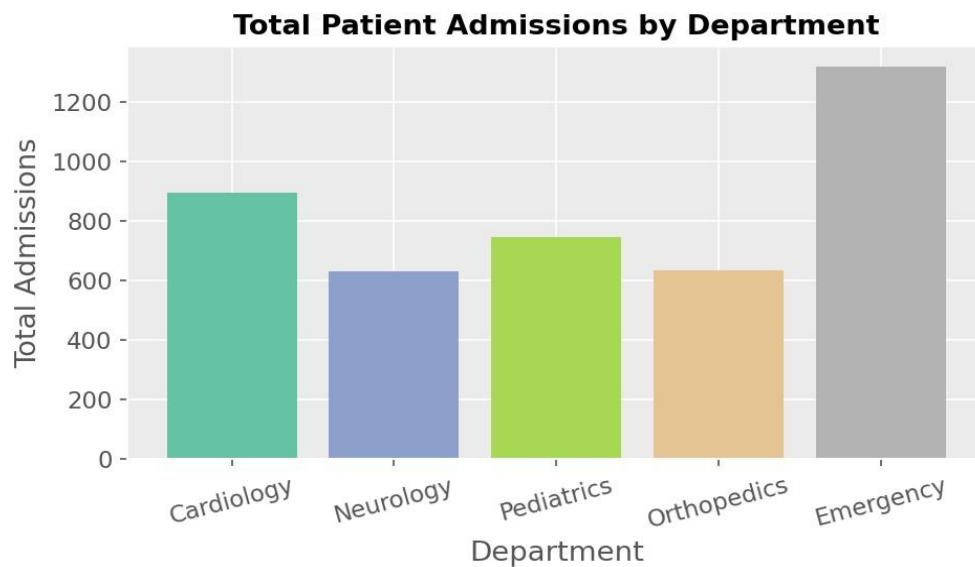
Set 1

Hospital Patient Admission Analysis

1. Create a bar chart to compare total patient admissions across departments. Label the axes and title the chart.

R code:

```
ggplot(hospital, aes(x = Department, y = Total_Admissions, fill = Department)) +  
geom_bar(stat = "identity") + labs(title = "Total Patient Admissions by Department", x = "Department", y = "Total Admissions") +  
theme_minimal() +  
theme(legend.position = "none")
```

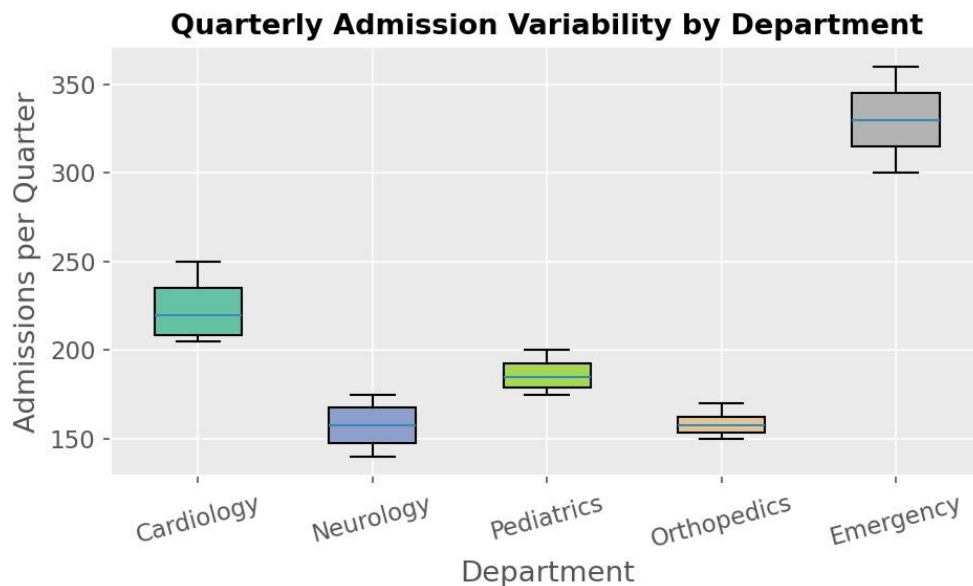


Output: Bar chart of total admissions by department

2. Create a box plot to evaluate the quarterly admission variability within each department.

R code:

```
ggplot(hospital_long, aes(x = Department, y = Admissions, fill = Department)) +  
geom_boxplot() + labs(title = "Quarterly Admission Variability by Department", x = "Department", y = "Admissions per Quarter") +  
theme_minimal() +  
theme(legend.position = "none")
```



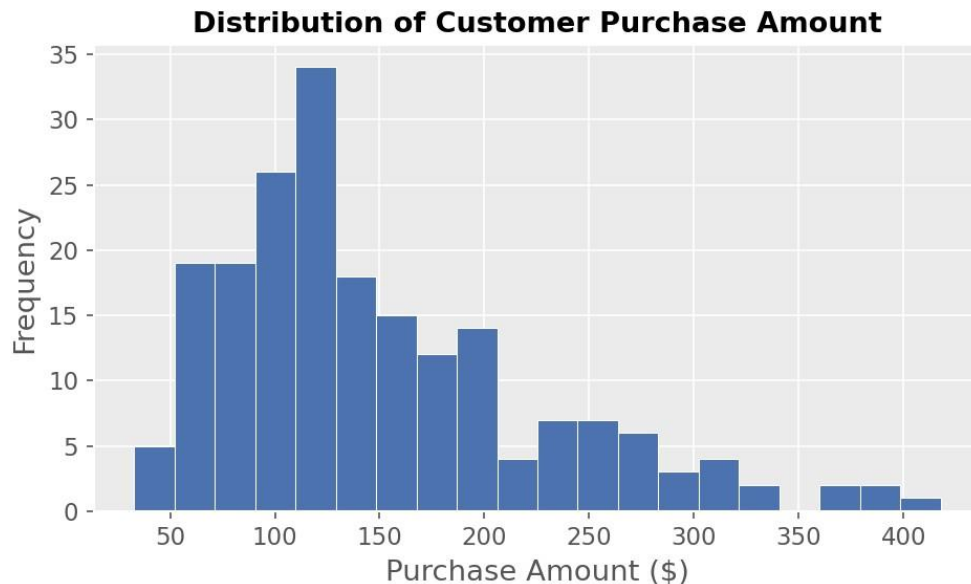
Output: Box plot of quarterly admission variability

Set 2: E-Commerce Customer Spending Behaviour

1. Create a histogram to examine the distribution of customer purchase amounts.

R code:

```
ggplot(ecommerce, aes(x = Purchase_Amount)) +
  geom_histogram(binwidth = 25, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Customer Purchase Amount", x =
    "Purchase Amount ($)", y = "Frequency") + theme_minimal()
```



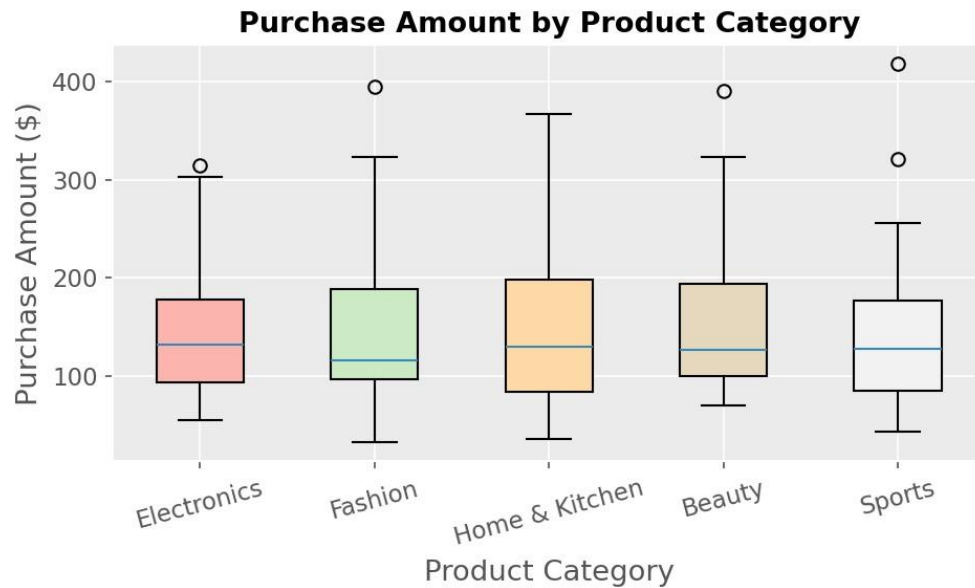
Output: Histogram of purchase amount

2. Create a box plot to compare purchase amounts across product categories and identify high-value customers.

R code:

```
ggplot(ecommerce, aes(x = Product_Category,
  y = Purchase_Amount, fill =
  Product_Category)) + geom_boxplot() + labs(
  title = "Purchase Amount by Product Category",
  x = "Product Category", y = "Purchase Amount
  ($)")
```

```
) + theme_minimal() +  
theme(legend.position = "none")
```



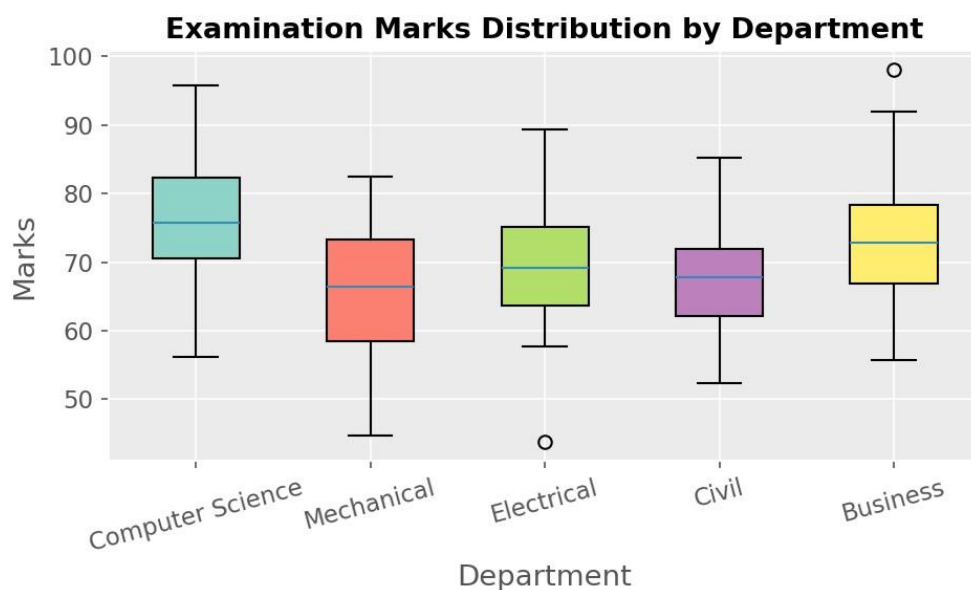
Output: Box plot of purchase amount by category

Set 3: University Examination Performance Dashboard

1. Create a box plot to compare score distributions across departments and identify outliers.

R code:

```
ggplot(exam, aes(x  
= Department, y  
= Marks, fill = Department)) + geom_boxplot() + labs(  
title = "Examination Marks Distribution by Department",  
x = "Department", y = "Marks" ) + theme_minimal() +  
theme(legend.position = "none")
```



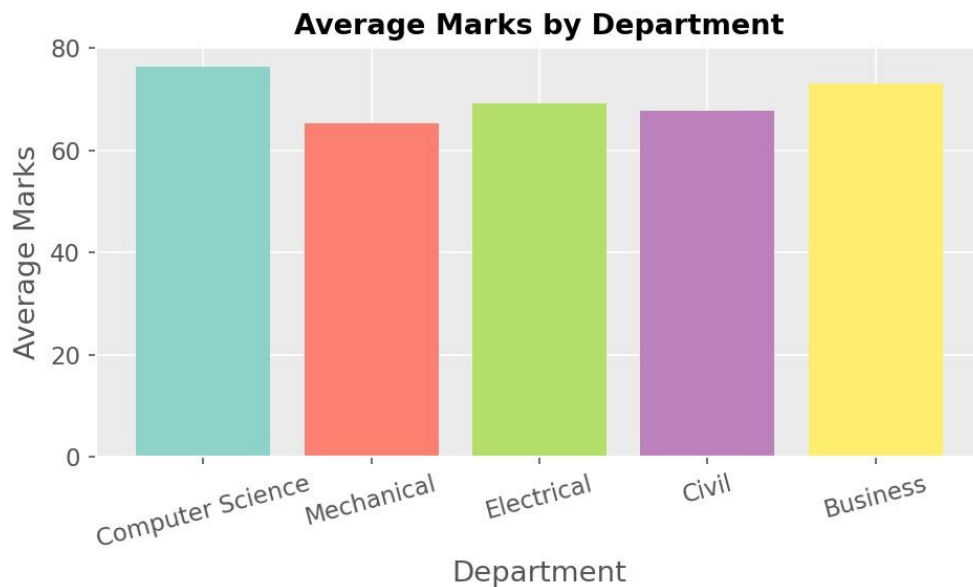
Output: Box plot of marks by department

2. Create a bar chart to compare the average marks across departments for academic decision-making.

R code:

```
exam_avg <- exam %>%
  group_by(Department) %>%
  summarise(Avg_Marks = mean(Marks))

ggplot(exam_avg,
  aes(x = Department,
      y = Avg_Marks, fill = Department))
+ geom_bar(stat = "identity") + labs(title =
"Average Marks by Department", x = "Department",
y = "Average Marks") + theme_minimal() +
  theme(legend.position = "none")
```



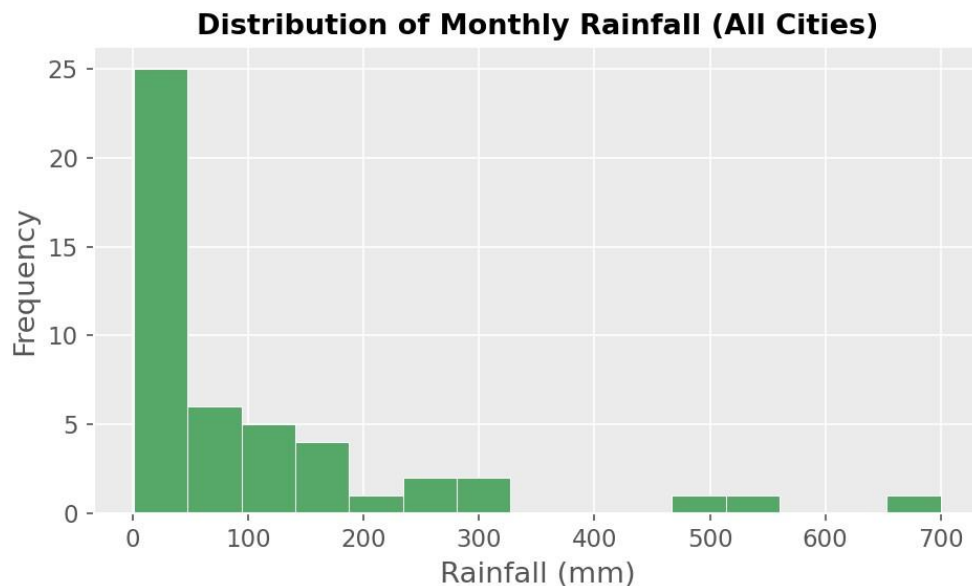
Output: Bar chart of average marks by department

Set 4: Climate and Rainfall Distribution Study

1. Create a histogram to represent the overall distribution of monthly rainfall across all cities.

R code:

```
ggplot(climate, aes(x = Rainfall_mm)) +
  geom_histogram(binwidth = 50, fill = "seagreen", color = "white") +
  labs(title = "Distribution of Monthly Rainfall (All Cities)",
x = "Rainfall (mm)", y = "Frequency") + theme_minimal()
```

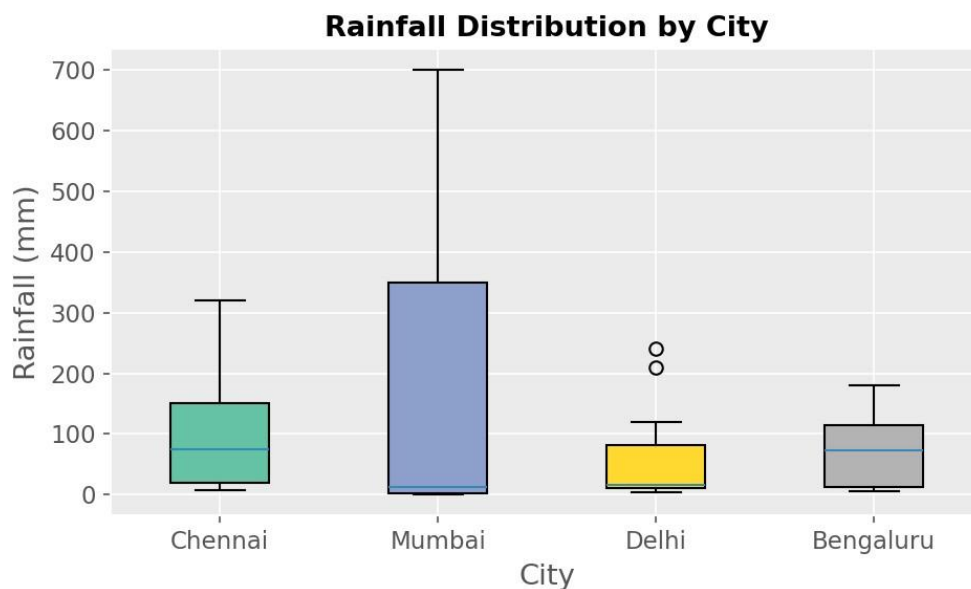


Output: Histogram of monthly rainfall

2. Create a box plot to compare rainfall distribution across cities and evaluate seasonal climate variability.

R code:

```
ggplot(climate, aes(x = City,
  y = Rainfall_mm, fill = City)) +
  geom_boxplot() + labs(title =
    "Rainfall Distribution by City", x =
    "City", y = "Rainfall (mm)" ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Output: Box plot of rainfall by city

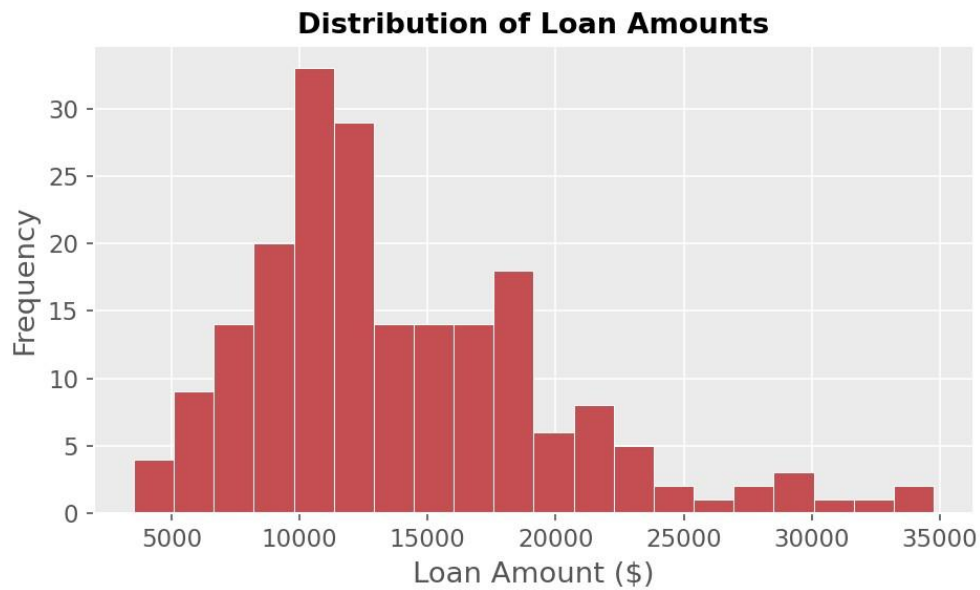
Set 5: Bank Loan Risk Assessment

1. Create a histogram to visualize the distribution of loan amounts.

R code:

```
ggplot(loans, aes(x = Loan_Amount)) + geom_histogram(binwidth =
  2000, fill = "indianred", color = "white") + labs(title =
```

```
"Distribution of Loan Amounts", x = "Loan Amount ($)", y =  
"Frequency" ) + theme_minimal()
```

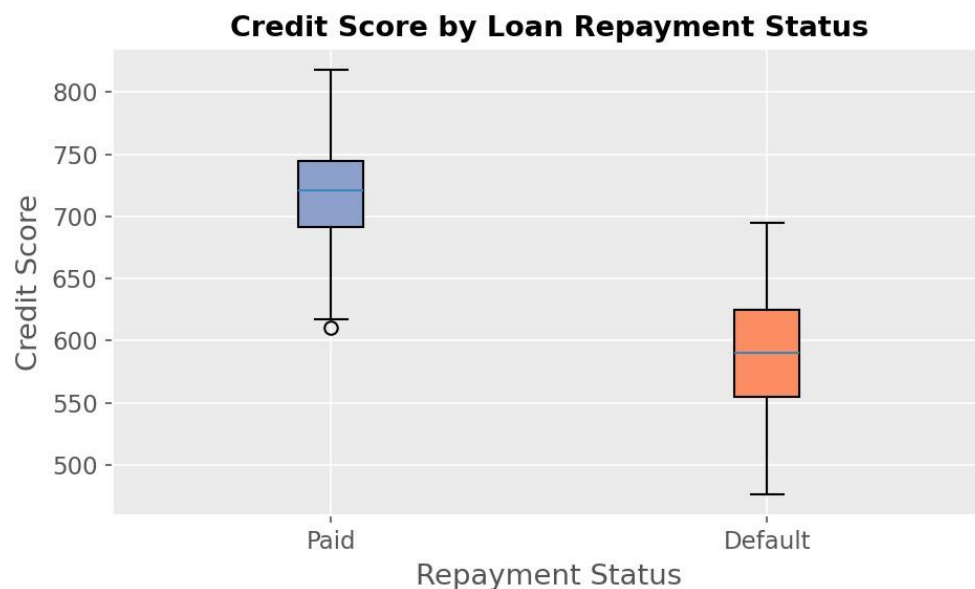


Output: Histogram of loan amounts

2. Create a box plot to compare credit scores between customers who repaid their loan and those who defaulted, to highlight risk patterns.

R code:

```
ggplot(loans, aes(x = Repayment_Status,  
y = Credit_Score, fill =  
Repayment_Status)) + geom_boxplot() + labs(  
title = "Credit Score by Loan Repayment Status",  
x = "Repayment Status", y = "Credit Score"  
) + theme_minimal() +  
theme(legend.position = "none")
```



Output: Box plot of credit score by repayment status